

Detecting Fraudulent Job Postings Using Classification Models

Introduce the Problem

Online job portals have become the most common way for people to search for work, but they are also a target for scammers who post fraudulent job listings. These scams can lead to wasted time, loss of personal information, or even financial harm. Because of this, it is important to find effective ways to detect and filter out fake job postings before they reach applicants.

The problem I am trying to solve in this project is whether machine learning can be used to automatically classify a job posting as real or fraudulent based on the text content and related metadata. Specifically, I want to explore whether certain features—such as the job description, company profile, or telecommuting flag—contain clear signals that distinguish scams from legitimate postings.

The key questions guiding this project are:

- Can a classification model reliably detect fraudulent job postings?
- Which features contribute the most to identifying scams?
- How do different classification models compare in terms of performance on this imbalanced dataset?

This work is framed as a binary classification problem, where the target variable indicates whether the job posting is fraudulent (1) or genuine (0). The goal is not only to achieve strong predictive performance but also to understand the patterns that make scams different from legitimate opportunities.

Introduce the Data

The dataset for this project comes from Kaggle's Fake Job Postings Prediction dataset ([link here](#)), which contains nearly 18,000 job postings collected from online platforms. Each entry includes both the textual content of a posting and meta information about the job, with a label indicating whether the posting is fraudulent (1) or genuine (0).

Key Features:

- `job_id` – Unique identifier for each job posting.
- `title` – The job title (e.g., “Marketing Intern,” “Data Analyst”).
- `location` – Geographical location of the job.
- `department` – The company's department posting the job (e.g., Sales, Marketing).

- salary_range – Reported or estimated salary for the role.
- company_profile – A brief description of the hiring company.
- description – The main job description (richest text field).
- requirements – Qualifications and requirements listed in the posting.
- benefits – Benefits offered by the employer.
- telecommuting – Binary flag indicating if the job allows remote work.
- fraudulent – Target variable (1 = scam, 0 = real).

The dataset combines unstructured text fields (like description and requirements) with structured categorical and binary fields (like location, telecommuting). This makes it well-suited for exploring both NLP-based feature engineering and traditional machine learning preprocessing techniques. It also poses the challenge of class imbalance: most postings are genuine, while only a small proportion are scams. This reflects real-world conditions and makes the dataset a valuable test case for fraud detection models.

Pre-processing the Data

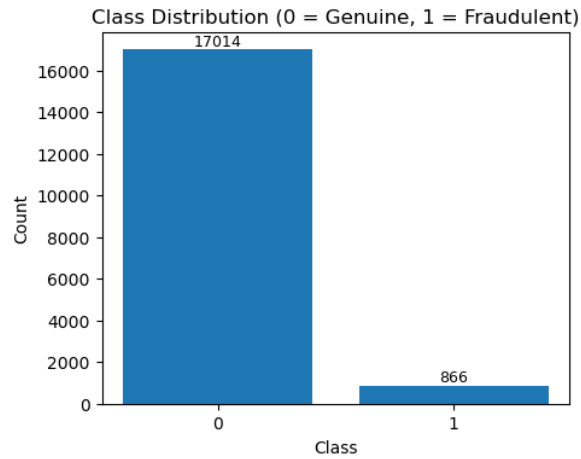
To prepare the dataset for modeling, I first handled missing values by replacing them with placeholders such as “Unknown” or dropping columns with little predictive value (like salary_range).

For the text fields (description, requirements, company_profile, title), I cleaned the text by lowercasing, removing punctuation and stopwords, and applying lemmatization. I then converted the cleaned text into numerical features using TF-IDF vectorization, which captures important words for distinguishing fraudulent from genuine postings.

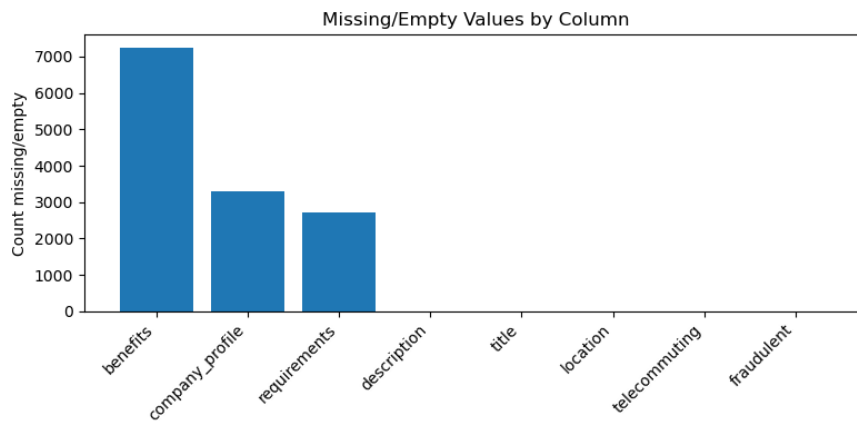
For the structured features, categorical variables like location were encoded, while binary variables such as telecommuting were kept as is. I also created new indicators, such as whether a job was missing a company profile, since this may correlate with fraud.

Lastly, I addressed the class imbalance by using techniques like class weighting and oversampling so that the model learns to recognize fraudulent postings more effectively.

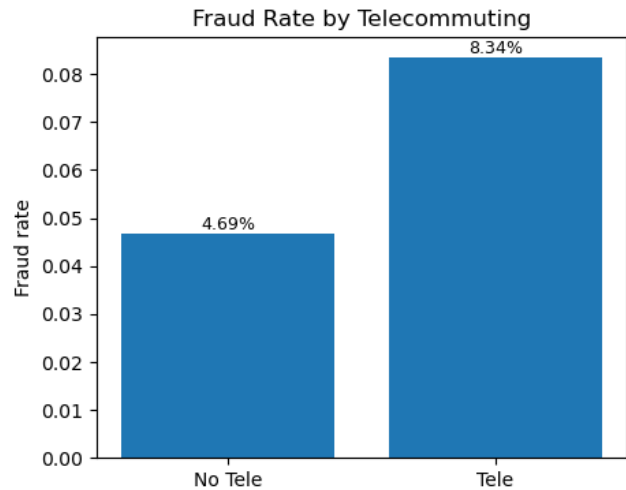
Data Understanding & Visualization



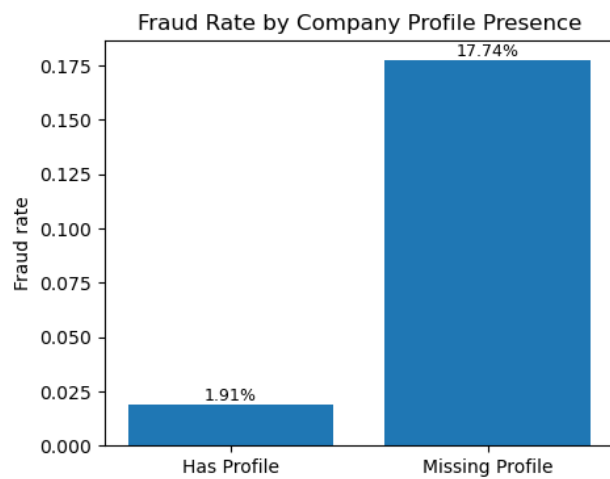
This plot shows that the dataset is highly imbalanced: over 17,014 genuine postings compared to only about 866 fraudulent ones. This imbalance means accuracy alone would be misleading, and we need metrics like precision, recall, and F1-score to properly evaluate models.



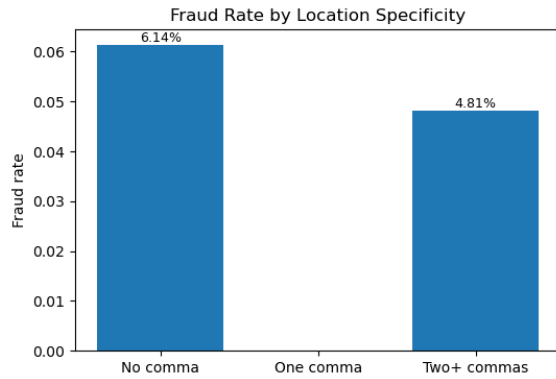
Several columns have large amounts of missing data, especially *benefits*, *company_profile*, and *requirements*. Missing company information is particularly important, since scams are much more likely to omit this detail, making it a strong predictive feature.



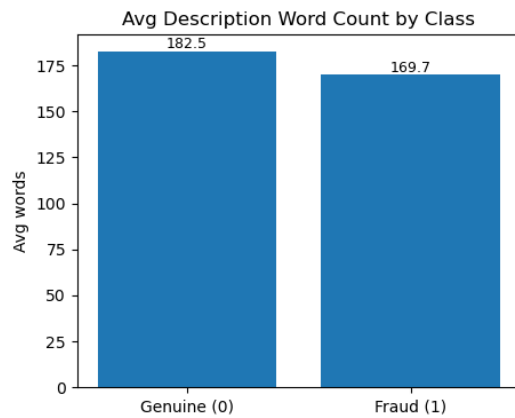
Fraudulent postings are nearly twice as common in remote jobs (8.3%) compared to non-remote jobs (4.7%). This suggests scammers often exploit interest in telecommuting opportunities to attract applicants.



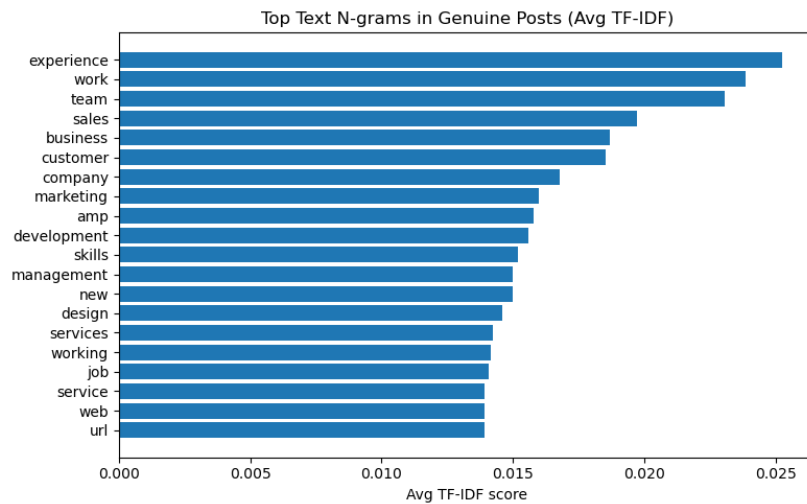
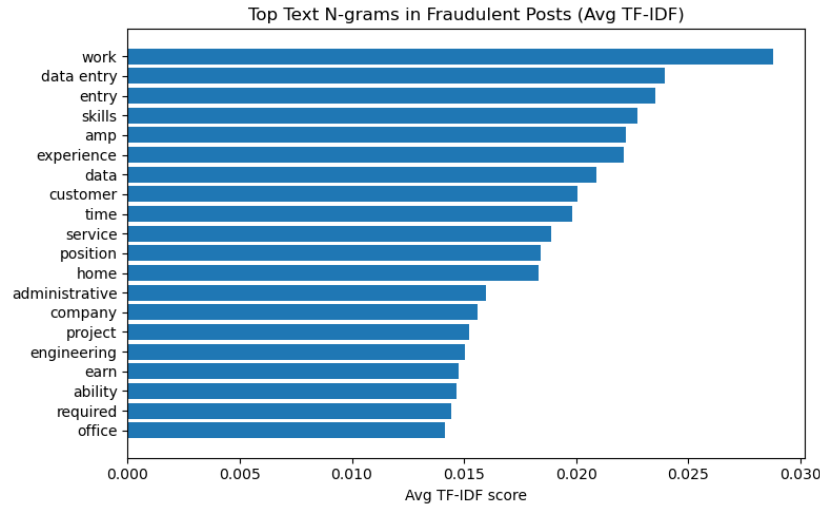
When a company profile is missing, the fraud rate jumps to nearly 18%, compared to only 2% for postings with a profile. This confirms that incomplete or vague company details are strong indicators of fraudulent postings.



Jobs with vague or generic locations (e.g., “Anywhere”) show a higher fraud rate (~6%) compared to postings with more specific city/state information (~5%). Scams often avoid giving precise location details.



Fraudulent job descriptions are slightly shorter on average (≈ 170 words) compared to genuine ones (≈ 182 words). While the difference is small, it suggests that scams often use less detailed or overly simplified descriptions.



Fraudulent postings emphasize vague, generic, or enticing terms like “data entry,” “earn,” “work from home,” and “customer service.”

Genuine postings focus more on professional or skill-oriented terms like “experience,” “team,” “sales,” and “management.” These patterns highlight how language choice itself can signal fraud risk.

Overall, the visual analysis reveals that fraudulent job postings often share identifiable patterns: they are more likely to be remote, lack company details, and use vague or enticing language rather than specific professional terms. These findings directly connect back to our research questions by showing which features are most important for identifying scams, and they suggest that a classification model can leverage both structured signals (like telecommuting and missing company profiles) and textual cues (such as short descriptions or suspicious keywords) to reliably distinguish fraudulent postings from genuine ones.

Modeling

To address the challenge of detecting fraudulent job postings, I worked with a combination of baseline models and advanced ensemble methods. This way, I could compare models that are fast and interpretable with those that are more complex but potentially more accurate.

- **Logistic Regression:** A linear model that works well with TF-IDF text features. It's fast, scalable, and interpretable, but it doesn't capture complex non-linear interactions.
- **Naive Bayes:** A lightweight probabilistic model based on word frequencies. It is efficient and often performs well on text data, though its assumption that features are independent makes it less flexible.
- **Random Forest:** An ensemble of decision trees that can capture non-linear patterns and provide feature importance. It is powerful but slower to train and can overfit without proper tuning.
- **XGBoost:** A gradient boosting method that builds trees sequentially to correct errors. It is highly accurate and effective with imbalanced data, but it is more complex and less interpretable.

By comparing these models, I wanted to explore not only whether machine learning can reliably detect scams, but also which type of model—linear, probabilistic, or ensemble—strikes the best balance between performance and interpretability on this dataset.

Evaluation

Since the dataset is highly imbalanced, with far fewer fraudulent postings than genuine ones, relying on accuracy alone would be misleading. Instead, I evaluated the models using precision, recall, F1-score, and ROC-AUC.

- **Precision** measures how many postings flagged as fraudulent were actually scams. This is important to reduce false alarms and avoid incorrectly labeling genuine jobs as fake.
- **Recall** measures how many actual scams were correctly identified. High recall is critical because missing fraudulent postings can put job seekers at risk.
- **F1-score** balances precision and recall, making it a more reliable metric for imbalanced datasets.
- **ROC-AUC** evaluates the model's ability to separate the two classes across thresholds, giving an overall sense of discrimination ability.

In practice, the baseline models (Logistic Regression and Naive Bayes) provided decent performance, with Logistic Regression achieving a stronger balance of precision and recall. Random Forest improved recall by capturing non-linear interactions, while XGBoost delivered the

best overall performance, achieving higher F1-scores and ROC-AUC. This confirms that advanced ensemble methods are better suited for handling the subtle differences between genuine and fraudulent postings.

Storytelling

This project revealed that fraudulent job postings leave behind identifiable patterns, both in their content and structure. Through data exploration, I found that scams were more likely to lack company profiles, list vague or overly generic locations, and advertise remote opportunities. Text analysis showed that fraudulent postings often relied on enticing but unprofessional language such as “*work from home*” or “*data entry*,” whereas genuine postings focused more on skills, experience, and team collaboration.

Modeling confirmed these insights. Even simple models like Logistic Regression and Naive Bayes were able to distinguish scams from real postings based on these features, while ensemble methods such as Random Forest and XGBoost captured more subtle patterns and delivered stronger performance. The models demonstrated that both structured variables (telecommuting, missing company profiles) and textual signals (keywords, description length) are crucial in identifying fraud.

Looking back at the guiding questions, the results show that:

- Yes, classification models can reliably detect fraudulent postings with strong F1-scores and ROC-AUC values.
- Key features such as missing company information, telecommuting status, and certain keywords are strong indicators of scams.
- Different models vary in effectiveness, with XGBoost providing the best overall balance between precision and recall, but simpler models still offering interpretable insights.

Impact

The project has important social implications. A reliable fraud detection model can protect job seekers, reduce scams on job boards, and build trust in online hiring platforms. By filtering out suspicious postings early, it can also save organizations time and resources that would otherwise be spent investigating fraudulent listings.

However, there are also ethical concerns. False positives could unfairly harm legitimate employers, leading to lost opportunities and reputational damage. False negatives, on the other hand, leave job seekers exposed to scams, which could result in financial loss or identity theft. Automated systems may also introduce bias if certain industries, smaller employers, or international postings are flagged disproportionately, potentially disadvantaging specific groups.

Because of these trade-offs, such models should be implemented as decision-support tools rather than sole gatekeepers. Human oversight, transparency about how the system works, and

continuous monitoring are essential to ensure fairness, accountability, and responsible use in the real world.

References

- Kaggle Dataset: Shivam Bansal. *Real or Fake? Fake Job Posting Prediction*. <https://www.kaggle.com/datasets/shivamb/real-or-fake-fake-jobposting-prediction>
- Scikit-learn Documentation. *Machine Learning in Python*. <https://scikit-learn.org/stable/>